ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

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Abstract. Adapting vision–language models (VLMs) such as Vintern-1B to a new Vietnamese Visual Question Answering (VQA) benchmark usually requires fine-tuning on many high-resolution image tiles or training a new architecture. We propose **ViBridge-VQA**, which improves Vintern-1B without updating any of its pre-trained weights. From a single image view, the frozen decoder receives *global* tokens, produced from the `[CLS]` vector by a small trainable bridge, and *local* tokens from Vintern-1B's own pre-aligned projector, pooled to a budget chosen as the knee of the accuracy–cost Pareto front. On the AutoViVQA benchmark, ViBridge-VQA trains 1.35% of the parameters, feeds its decoder 36 times fewer visual tokens than the official fine-tuning and reaches 54.6 F1 on an image-disjoint split; a lightweight decoder LoRA raises F1 to 58.2. Reusing the pre-trained projector outperforms learned connectors, and the local tokens also help on two other Vietnamese VQA datasets.

Keywords: Vietnamese Visual Question Answering · Vision–Language Models · Parameter-Efficient Learning · Visual Token Budget · Generative VQA

1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image. Large vision–language models (VLMs) such as BLIP-2 [13], Flamingo [1] and LLaVA [20] generate free-form answers by connecting a pre-trained vision encoder to a pre-trained language model. For Vietnamese, the open Vintern-1B [8] connects an InternViT-300M encoder [7] to a Qwen2.5-0.5B decoder [27], and benchmarks such as ViVQA [30] and the reasoning-oriented AutoViVQA [22] measure progress in this low-resource setting.

Adapting such a model to a new benchmark remains expensive. The fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA adapters to the language model

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and splits each test image into six high-resolution tiles plus a thumbnail, so its decoder reads 1,792 visual tokens per image. ViMoE-VQA [21] instead designs a task-specific mixture-of-experts fusion module. A cheaper option keeps the VLM frozen and trains only a small connector, usually of one of two designs. A *global-only bridge* maps the global image vector to a few tokens: it is cheap but loses small objects, printed text and spatial relations. A *learned patch compressor*, such as a Q-Former, keeps these details but must learn a new mapping to the decoder, which is hard with limited Vietnamese data.

Vintern-1B offers a third option. Its projector is already pre-trained to map image regions into the input space of the decoder, so this alignment need not be learned again. The model also discards the global `[CLS]` vector of its encoder, which summarises the whole scene. We therefore ask whether, with limited labelled data, reusing this alignment beats learning a new connector, and how many global and local tokens the frozen decoder needs.

We propose **ViBridge-VQA**, which feeds the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens reuse the frozen projector, average-pooled to a chosen budget and placed in the original image slot, so the decoder reads region features in the representation it was trained on. *Global* tokens, produced from the `[CLS]` vector by a small trainable bridge, add the scene-level information that Vintern-1B ignores. The bridge is the only trained part, so Vintern-1B stays unchanged and can be shared across tasks. As every token adds inference cost, we treat the number of tokens as a budget and select the knee of the accuracy–cost Pareto front. Without updating any weight of Vintern-1B, ViBridge-VQA reaches an F1 close to that reported for its fine-tuned version, while feeding its decoder 36 times fewer visual tokens. The key is pre-alignment: at a similar number of tokens, global tokens plus the projector's local tokens beat a learned Q-Former by 6.0 F1 and a C-Abstractor by 2.2 F1. Our contributions are:

- **ViBridge-VQA**, a global–local bridge that combines trainable global tokens with local tokens from the frozen projector of the VLM, optionally with a lightweight decoder LoRA.
- An explicit **accuracy–cost budget** for the number of visual tokens, selected on validation as the knee of the Pareto front.
- An **analysis of where the gains come from**: reusing the pre-aligned projector beats learned connectors with up to 9.4× more parameters, local tokens matter more than global ones, and they also help on two other Vietnamese VQA datasets.

2 Related Work

Vietnamese VQA. Vietnamese VQA datasets include ViVQA [30], the synthetic ViCLEVR [32], the book-cover ViOCRVQA [26] and the scene-text ViTextVQA [24]. AutoViVQA [22] contributes 37,077 questions on 19,411 COCO images [18], generated by large language models, filtered by an ensemble of validators and labelled with reasoning types. On the modelling side, BARTPhoBEiT [31]

and AViVQA-TranConI [23] combine pre-trained visual and Vietnamese language representations, Vintern-1B [8] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [21] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA. AutoViVQA and ViMoE-VQA are earlier work of our group: we use the AutoViVQA data but no component of ViMoE-VQA, and the method, the image-disjoint split and all results are new. These approaches train a new model or fine-tune a VLM on multi-tile inputs, whereas we keep Vintern-1B frozen and use a single image view.

Connectors. Connectors between a frozen vision encoder and a frozen language model differ mainly in what they must learn. Frozen [33] learns a visual prefix from a global image representation. The Q-Former of BLIP-2 [13] learns queries that compress the patch features: it preserves details but is a large module (188M parameters) pre-trained on millions of image–text pairs. LLaVA [20,19] projects every patch token with a small MLP, which preserves locality but passes hundreds of tokens to the decoder. Honeybee [4] proposes locality-preserving abstractors, and InternVL [7], the basis of Vintern-1B, merges neighbouring patches before its MLP projector. In all of these models, the mapping from patches to the decoder is learned during large-scale pre-training. ViBridge-VQA does not learn it again: it reuses the pre-trained projector and trains only a small global bridge (12.9M parameters) on about 26k task questions.

Visual token budget. The number of visual tokens largely determines the cost of the decoder. Inference-optimal scaling of VLMs [14] shows that a few tokens can suffice for many tasks, and Matryoshka multimodal models [3] show with nested pooled budgets that the required budget depends on the task. Other work reduces the tokens directly: DeCo [36] replaces the Q-Former with adaptive average pooling of the patch tokens, TokenPacker [15] learns a compact projector, LLaVA-PruMerge [29] and FastV [5] prune or merge the tokens of an existing model, and VisionZip [35] keeps those that receive the most attention from `[CLS]`. Like DeCo, we use average pooling, but on the output of an already pre-trained projector with the whole VLM frozen, and we choose the budget with explicit criteria: the knee of the Pareto front [28] and TOPSIS [12]. Parameter-efficient methods such as adapters [10], prefix tuning [16] and LoRA [11] reduce the cost of adaptation from the other side, by training a few additional weights in the decoder. We use decoder LoRA both as a baseline and as a complement to ViBridge-VQA.

3 Proposed Method

We formulate VQA as conditional answer generation: given an image I and a Vietnamese question q, the model generates a short free-form answer $y = (y_1, \ldots, y_T)$. ViBridge-VQA keeps every pre-trained weight of Vintern-1B [8] frozen and changes only the visual input of its decoder.

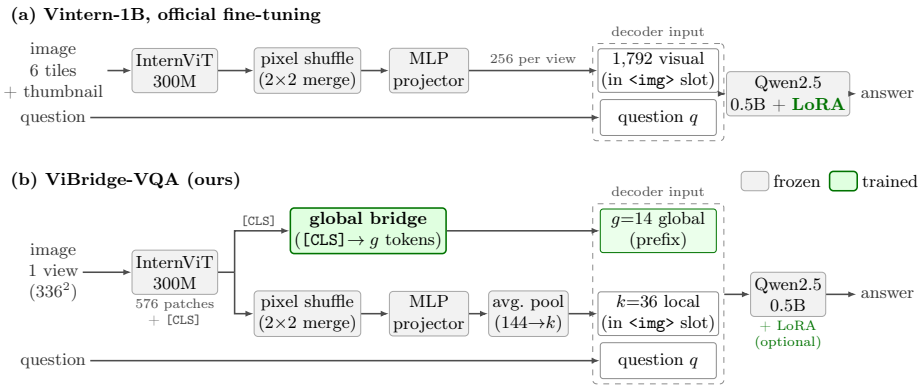


Fig. 1: Overview. (a) Official fine-tuning of Vintern-1B, with six tiles plus a thumbnail for every AutoViVQA test image. (b) ViBridge-VQA on one view: k local tokens, average-pooled from the 144 region tokens of the frozen projector, fill the image slot, and g global tokens from `[CLS]` precede the prompt. The selected budget is $g = 14$, $k = 36$ (50 tokens). The decoder stays frozen; a LoRA is optional.

3.1 Architecture

As shown in Fig. 1, ViBridge-VQA keeps the three components of Vintern-1B: the InternViT-300M vision encoder (24 layers, width 1024), its two-layer MLP projector P and the Qwen2.5-0.5B-Instruct decoder (24 layers, width 896). Each image is resized to one 336×336 view, i.e., 24×24 patches of 14×14 pixels. InternViT, pre-trained at 448×448, bicubically interpolates its position embeddings to this resolution. The encoder returns one vector per patch and a global `[CLS]` vector $v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: pixel shuffle [6] merges each 2×2 block of patches into one 4096-dimensional vector, and P, pre-trained to align the two backbones, maps it into the decoder input space. This yields a 12×12 grid of 144 region tokens. ViBridge-VQA uses both outputs of the encoder.

3.2 Global Tokens

The `[CLS]` vector summarises the whole scene, but Vintern-1B never passes it to the decoder. A small trainable bridge f_ϕ expands it into g tokens with two linear heads:

$$f_\phi(v) = \left[\, W_a v + b_a;\ \mathrm{reshape}_{(g-1)\times 896}(W_e v + b_e) \,\right] \in \mathbb{R}^{g \times 896}, \tag{1}$$

where $W_a \in \mathbb{R}^{896\times 1024}$ and $W_e \in \mathbb{R}^{(g-1)896\times 1024}$. The first head gives an *anchor* token, and the second gives $g - 1$ complementary tokens that present the global content along different learned directions. The bridge has $1025 \cdot 896 \cdot g$ parameters (12,857,600 for $g = 14$), and its tokens are placed before the prompt. When the bridge alone is varied from 4 to 18 tokens, F1 peaks at $g = 14$ (Section 4.5), the value used in ViBridge-VQA. Our Global-only baseline uses $g = 8$, matching the eight output tokens of Tile-Attention and the light Q-Former (Section 4.4).

3.3 Local Tokens

To give the decoder region-level detail, ViBridge-VQA also passes local tokens taken directly from the 144 region tokens of P. We average-pool the 12×12 grid to an $s \times s$ grid, so $k = s^2$ with $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, and $k = 36$ averages each 2×2 block. The pooled tokens replace the embeddings of k placeholders inside Vintern-1B's own image slot, `<img>` $\cdots$ `</img>`. Because they come from the pre-trained projector rather than a new projection, they stay in the representation that the decoder was trained to read, and they need no parameters.

3.4 Training

By default, only the bridge parameters ϕ are trained; the encoder, the projector and the decoder stay frozen. We minimise the cross-entropy of the answer tokens:

$$\mathcal{L} = -\sum_{t=1}^{T} \log p\big(y_t \mid y_{<t},\, f_\phi(v),\, \mathrm{pool}_s(P(x)),\, q\big), \tag{2}$$

where x denotes the merged patch vectors. Optionally, ViBridge-VQA is combined with decoder adaptation: a rank-16 LoRA [11] on the query, key, value and output projections of every decoder layer, trained jointly with the bridge, adds 2,162,688 parameters. Percentages of trained parameters are relative to all model parameters (the frozen 0.94B plus the trained ones). With $g = 14$, the bridge trains 12.86M parameters (1.35%), and the bridge with LoRA 15.02M (1.57%).

3.5 Budget Selection

The cost of the decoder grows with the number of visual tokens $g + k$, so we treat the pair (g, k) as a budget. Among the configurations with local tokens, F1 increases with every added token in our experiments, so all of them lie on the accuracy–token Pareto front, which alone does not single one out. We therefore select its *knee* [28], after which additional tokens bring diminishing returns. With accuracy a the validation F1 and cost c the number of visual tokens, both normalised to $[0, 1]$ over the front, the knee maximises $\hat{a} - \hat{c}$, i.e., it lies furthest above the line joining the cheapest and the most accurate configuration. We check the choice with TOPSIS [12], which weighs accuracy (F1, CIDEr, METEOR) and cost (tokens, evaluation time, trained parameters, seed variance) equally. Both criteria use validation only, and the code with all rankings is released with the paper.

4 Experiments

4.1 Dataset and Metrics

Dataset. We use AutoViVQA [22], built on MS COCO images [18], where each question has five short free-form answers and a reasoning type. After removing

273 duplicated (image, question) pairs and 97 questions without a reasoning label, we split the remaining 36,707 questions 70/15/15 by image (seed 42), so no test image is seen in training: 25,776 / 5,463 / 5,468 questions on 13,579 / 2,909 / 2,910 images. This split is stricter than a question-level one, on which Global-only scores about 0.5 F1 higher.

Metrics. Following AutoViVQA and ViMoE-VQA, we report Accuracy (Acc), word-level Precision (Prec), Recall (Rec) and F1, BLEU [25], ROUGE-L (R-L) [17], METEOR (MET) [2] and CIDEr [34] against the five references. Precision, Recall and F1 compare the word sets of the answer and of the reference with the highest F1, and are averaged over questions. Accuracy is the share of answers identical to a reference, ignoring case; BLEU is corpus-level BLEU-4 on whitespace tokens.

4.2 Setup

Implementation details. The encoder, projector and decoder come from Vintern-1B-v3.5 [8] and stay frozen. We train with AdamW (learning rate 2×10^{-4}, effective batch size 8) on the first reference answer. Bridges are trained for two epochs, after which the validation loss and CIDEr stop improving. Models with a decoder LoRA are trained for one epoch, with the bridge (if any) and the LoRA learned jointly from scratch. Global-only + LoRA is also reported after three epochs, its best setting, which Table 6 and Fig. 3 use. Decoding is greedy with at most 50 new tokens, and every model fits on one 16 GB NVIDIA T4. Each configuration is trained with three seeds (42, 123, 3407), four for Global-only (adding 2026), and tables report mean ± std.

Baselines. Table 1 compares with two groups of models: (1) fine-tuned on AutoViVQA: ViT5_ViT, BARTPhoBEiT [31], Vintern-1B and ViMoE-VQA [21], and (2) zero-shot: Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5. The fine-tuned Vintern-1B follows the official recipe: six 448×448 tiles plus a thumbnail per test image and a rank-16 LoRA on the language model for one epoch. The analysis also uses *Global-only*, the bridge of Eq. (1) with $g = 8$ and no local tokens, with and without a decoder LoRA. Our models are *ViBridge-VQA* ($g = 14$, $k = 36$) and *ViBridge-VQA + LoRA*.

4.3 Main Results

ViBridge-VQA nears fine-tuned performance. Without updating any weight of Vintern-1B, ViBridge-VQA raises its F1 from 17.55 (zero-shot) to 54.56 (Table 1). This is comparable to the reported 53.76 of the fine-tuned Vintern-1B, on a stricter split, while the decoder receives 50 instead of 1,792 visual tokens and only 1.35% of the parameters are trained. Only Accuracy is lower (11.75 versus 13.01). On test, the bridge alone with the same 14 global tokens reaches 51.02 F1 (±0.10), so the 36 local tokens add 3.54 F1.

Table 1: Results on AutoViVQA. Baselines are as reported in AutoViVQA [22] and ViMoE-VQA [21]; ours are mean ± std over three seeds on the image-disjoint test split. In bold: the best baseline per column, or our results that exceed it. All values ×100.

Model	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
Fine-tuned								
ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
BARTPhoBEiT	8.81	45.30	46.48	45.88	4.33	44.83	24.57	**188.96**
Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
ViMoE-VQA	9.55	**62.89**	58.60	**60.69**	12.54	47.12	39.10	88.76
Zero-shot								
Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
Gemini 2.0 F.	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
Gemini 2.5 F.	0.22	24.43	**76.66**	24.75	0.39	37.27	31.22	71.90
GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Ours ($g = 14$, $k = 36$)								
ViBridge-VQA	11.75 ±.21	55.36 ±.21	56.56 ±.21	54.56 ±.19	**18.52** ±.18	**52.64** ±.19	**45.53** ±.33	110.58 ±.95
ViBridge-VQA + LoRA	**14.67** ±.30	59.02 ±.12	59.96 ±.33	58.17 ±.19	**22.53** ±.11	**56.25** ±.22	**49.32** ±.37	122.88 ±.93

LoRA adds 3.6 F1. With a decoder LoRA, ViBridge-VQA reaches 58.17 F1 (+3.61) with 1.57% of the parameters trained, and gives the best Accuracy (14.67), BLEU (22.53), ROUGE-L (56.25) and METEOR (49.32). ViMoE-VQA, which trains its fusion module and decoder on AutoViVQA, keeps the best Precision and F1. Results are stable across seeds (test std of F1 at most 0.19) and splits (test F1 within 0.4 of validation).

4.4 Token Sources

Table 2 compares three sources of visual tokens. *Residual* (one token: a linear projection plus a two-layer MLP correction) and *Global-only* (Eq. (1), $g = 8$) read the `[CLS]` vector. Five bridges learn to compress the 577 encoder tokens: *Tile-Attention* (self-attention and eight learned queries), the light and full Q-Former [13] (7 and 15 learned queries plus an anchor token from the mean encoder feature; two and four layers; the full one also reads the question), C-Abstractor [4] (convolutions around a 3×3 pooling, 9 tokens) and *Patch-pool* (a linear projection averaged into eight bands). All are randomly initialised and trained with the same recipe. The third source is our local tokens.

Learned compressors fall short. The five learned compressors stay 1.9–5.6 F1 below Global-only, despite up to 9.4× more parameters. With limited data, a new mapping from patches to the decoder is hard to learn, and more capacity does not help: the 69.4M-parameter full Q-Former is no better than the 27.6M light one.

Table 2: Sources of visual tokens with a frozen decoder, on validation. Below each name: number of visual tokens and trained parameters. Best trained model in bold. All values ×100.

Source	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
Residual 1 token, 4.86M	5.63 ±.59	46.31 ±.34	48.79 ±.33	46.02 ±.36	12.98 ±.15	44.23 ±.38	36.86 ±.31	86.48 ±.77
Global-only 8 tokens, 7.35M	9.21 ±.05	51.54 ±.15	52.48 ±.21	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
Tile-Attention 8 tokens, 4.14M	4.99 ±1.10	45.49 ±1.10	48.05 ±.60	45.14 ±.92	12.34 ±.32	43.28 ±.95	36.11 ±.73	84.14 ±1.40
Light Q-Former 8 tokens, 27.6M	6.94 ±.32	47.74 ±.80	49.18 ±.51	47.06 ±.66	13.99 ±.33	45.37 ±.63	37.57 ±.60	88.07 ±1.89
Full Q-Former 16 tokens, 69.4M	6.97 ±.33	47.42 ±.70	49.02 ±.35	46.84 ±.47	13.76 ±.36	45.10 ±.41	37.43 ±.34	87.74 ±1.24
C-Abstractor 9 tokens, 19.87M	8.04 ±.07	49.41 ±.32	50.93 ±.32	48.86 ±.33	15.27 ±.13	47.17 ±.33	39.60 ±.33	94.05 ±1.32
Patch-pool 8 tokens, 0.92M	5.58 ±.27	46.11 ±.39	48.52 ±.49	45.78 ±.44	12.71 ±.49	43.99 ±.47	36.69 ±.46	85.83 ±1.43
Local only, untrained 36 tokens, no parameters	0.00	18.07	68.17	27.55	2.47	20.82	34.50	67.20
ViBridge-VQA 50 tokens, 12.86M	**12.38** ±.22	**55.86** ±.21	**56.72** ±.33	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01

Pre-alignment is the key. The same patch features help once they pass through the pre-trained projector. Adding 36 or 144 local tokens to Global-only ($g = 8$) raises F1 by 3.58 and 5.41 (Table 3). At a similar number of tokens, the projector wins with fewer trained parameters: 17 tokens ($g = 8$, $k = 9$) give 52.82 F1 against 46.84 for the 16-token full Q-Former, and 9 tokens ($g = 8$, $k = 1$) give 51.03 against 48.86 for the 9-token C-Abstractor. Pooling alone is not the reason: Patch-pool also averages, but after a new projection, and reaches only 45.78 F1. Distilling the projector into Global-only does not help either: 50.35 F1 with a cosine loss on the mean tokens and 50.46 with a KL loss on the answers, against 50.74 without distillation.

4.5 Token Budget

Local tokens matter more than global tokens. Table 3 and Fig. 2b vary $g \in \{8, 14\}$ and $k \in \{1, 9, 36, 144\}$. At $g = 14$, going from 1 to 144 local tokens adds 4.94 F1 and 19.05 CIDEr, whereas going from 8 to 14 global tokens adds only 0.2–0.8 F1 at a fixed k. The gain per token falls from 0.07 F1 (23 to 50 tokens) to 0.015 F1 (50 to 158 tokens).

The knee is $g = 14$, $k = 36$. The knee of the Pareto front (Section 3.5, Fig. 2b) is $g = 14$, $k = 36$ for each of the eight validation metrics, and also with generation time as the cost. TOPSIS with equal weights also ranks it first. $k = 144$ is 1.65 F1 higher, but $k = 36$ obtains 72% of its gain over Global-only (67% over the 14-token bridge of Table 4) with 32% of its tokens.

Global tokens peak at $g = 14$. Table 4 and Fig. 2a vary g from 4 to 18 for Global-only. Validation F1 rises from 48.98 ($g = 4$) to 50.74 ($g = 8$) and 51.56

Table 3: Effect of the token budget, on validation. Time: minutes to generate all validation answers on one T4. Selected budget in bold. All values ×100.

g	k	Tokens	Time	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
8	0	8	33.9	9.21 ±.05	51.54 ±.15	52.48 ±.21	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
14	0	14	37.2	9.74 ±.21	52.47 ±.11	53.16 ±.14	51.56 ±.08	17.76 ±.01	49.86 ±.05	42.14 ±.17	101.18 ±.52
8	1	9	34.2	9.33 ±.29	52.03 ±.12	52.75 ±.10	51.03 ±.12	16.54 ±.27	49.21 ±.13	41.37 ±.11	99.27 ±.58
8	9	17	35.5	10.25 ±.26	53.63 ±.04	54.72 ±.16	52.82 ±.11	17.58 ±.03	50.96 ±.05	43.56 ±.20	106.11 ±.55
8	36	44	36.0	12.15 ±.44	55.19 ±.43	56.14 ±.40	54.32 ±.41	18.22 ±.39	52.48 ±.43	45.31 ±.42	111.88 ±1.43
8	144	152	41.7	13.83 ±.12	56.94 ±.20	58.10 ±.15	56.15 ±.08	19.82 ±.00	54.21 ±.06	47.51 ±.05	119.05 ±.25
14	1	15	35.3	9.67 ±.14	52.50 ±.40	53.48 ±.14	51.67 ±.25	17.30 ±.36	49.93 ±.24	42.36 ±.19	101.66 ±1.11
14	9	23	34.7	10.55 ±.22	53.96 ±.10	54.66 ±.11	52.98 ±.08	17.83 ±.23	51.17 ±.10	43.71 ±.13	106.02 ±.80
14	**36**	**50**	**35.8**	**12.38** ±.22	**55.86** ±.21	**56.72** ±.33	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01
14	144	158	41.1	14.19 ±.36	57.36 ±.07	58.58 ±.03	56.61 ±.04	20.18 ±.13	54.70 ±.05	48.00 ±.05	120.71 ±.12

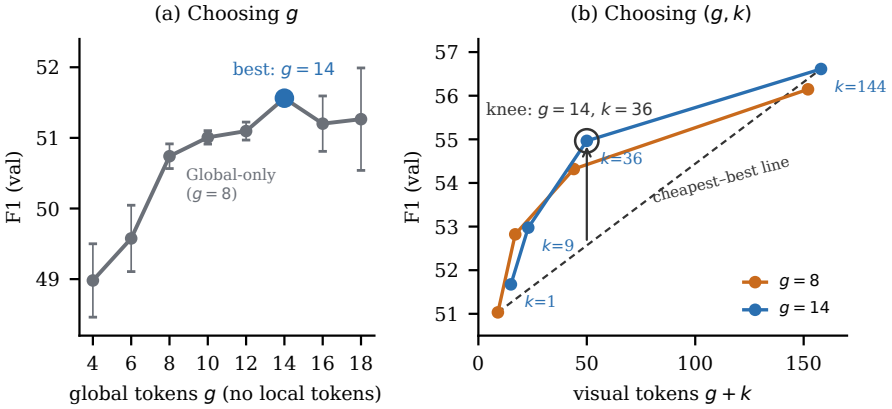


Fig. 2: Budget selection (validation F1). (a) Global-only peaks at $g = 14$. (b) With local tokens, the knee is the configuration furthest above the line joining the cheapest and the most accurate one.

($g = 14$), then falls to 51.20 and 51.26 at $g = 16$ and 18; CIDEr follows the same trend.

Fewer visual tokens, one encoder pass. Compared with the official tiling, ViBridge-VQA reduces the visual tokens at the decoder input from 1,792 to 50, and the encoder computation from seven 448 × 448 views (362 GMACs each) to one

Table 4: Global-only with g tokens, on validation. Selected g in bold. All values $\times 100$.

g	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
4	7.77 ±.27	49.57 ±.48	51.07 ±.57	48.98 ±.52	15.17 ±.48	47.23 ±.58	39.50 ±.59	93.56 ±1.42
6	8.38 ±.39	50.34 ±.53	51.43 ±.43	49.58 ±.47	15.67 ±.29	47.91 ±.48	40.09 ±.50	95.51 ±1.59
8	9.21 ±.05	51.54 ±.15	52.48 ±.21	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
10	9.24 ±.14	51.95 ±.06	52.58 ±.13	51.01 ±.10	17.03 ±.22	49.26 ±.11	41.46 ±.07	99.20 ±.25
12	9.41 ±.08	52.03 ±.12	52.64 ±.21	51.10 ±.13	17.06 ±.30	49.47 ±.08	41.52 ±.23	99.26 ±.54
14	**9.74** ±.21	**52.47** ±.11	**53.16** ±.14	**51.56** ±.08	**17.76** ±.01	**49.86** ±.05	**42.14** ±.17	**101.18** ±.52
16	9.44 ±.37	51.99 ±.58	52.91 ±.22	51.20 ±.39	17.29 ±.63	49.53 ±.36	41.88 ±.31	100.29 ±.95
18	9.45 ±.45	52.13 ±.72	52.88 ±.69	51.26 ±.73	17.41 ±.54	49.56 ±.68	41.81 ±.69	100.46 ±2.02

Table 5: Effect of a rank-16 decoder LoRA (2.16M parameters, one joint epoch), on test. Bridge parameters: 7.35M for Global-only, 12.86M for ViBridge-VQA. Best in bold. All values $\times 100$.

Model	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
Global-only	8.86 ±.37	51.19 ±.20	52.37 ±.21	50.48 ±.22	16.24 ±.43	48.70 ±.19	41.10 ±.29	96.20 ±1.10
Global-only + LoRA	10.47 ±.22	53.95 ±.04	54.72 ±.07	53.10 ±.05	19.00 ±.16	51.29 ±.08	43.63 ±.16	104.44 ±.45
ViBridge-VQA	11.75 ±.21	55.36 ±.21	56.56 ±.21	54.56 ±.19	18.52 ±.18	52.64 ±.19	45.53 ±.33	110.58 ±.95
ViBridge-VQA + LoRA	**14.67** ±.30	**59.02** ±.12	**59.96** ±.33	**58.17** ±.19	**22.53** ±.11	**56.25** ±.22	**49.32** ±.37	**122.88** ±.93

336×336 view (191 GMACs), i.e., 13 times fewer encoder GMACs. Within our budgets, $k = 144$ makes generation about 15% slower than $k = 36$.

4.6 Decoder Adaptation

LoRA gains more with local tokens. ViBridge-VQA + LoRA is the best model on every metric of Table 5 (58.17 F1, 122.88 CIDEr), and LoRA gains more on ViBridge-VQA than on Global-only for all eight metrics (+3.61 versus +2.62 F1, +12.31 versus +8.24 CIDEr). Without local tokens, LoRA lifts Global-only to 53.10 F1, below ViBridge-VQA with a frozen decoder (54.56), and only matches it after three epochs (54.52); Residual, Tile-Attention and the two Q-Formers with LoRA reach 52.69–53.29 validation F1. With a frozen decoder, a trained bridge is needed: local tokens alone give 27.55 F1, mostly because the decoder then answers in long sentences (24.8 words on average).

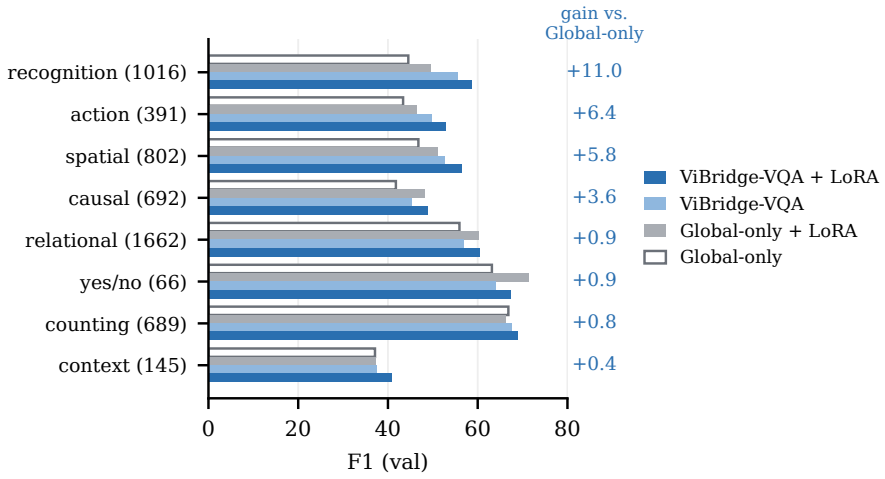


Fig. 3: Validation F1 by reasoning type (number of questions), sorted by the gain of ViBridge-VQA over Global-only (right column).

4.7 Analysis

Local tokens help most on visible content. In Fig. 3, ViBridge-VQA adds 11.0 F1 over Global-only on recognition, 6.4 on action and 5.8 on spatial questions, and beats Global-only + LoRA on these types without adapting the decoder. Decoder adaptation helps more on causal and relational questions, which have the longest reference answers (5.0 and 5.6 words), and on the 66 yes/no questions. ViBridge-VQA + LoRA is the best of the four models on seven of the eight types, all but yes/no.

More exact answers, fewer misses. On validation, answers are as long as the references (4.3 words on average), and most of them (77–81%) partially overlap the closest reference. Compared with Global-only, ViBridge-VQA raises exact matches (F1 = 1) from 9.7% to 13.3% and lowers answers with no word in common with any reference (F1 = 0) from 9.6% to 8.0%; with LoRA, these reach 15.7% and 6.8%.

LoRA makes answers precise. In Fig. 4, ViBridge-VQA already finds the relevant content, such as the driver's phone or the helmets, but its answers are vague. The decoder LoRA makes them precise: the driver is using *the phone*, and the helmets are worn *to protect their heads*.

4.8 Cross-Dataset Results

Table 6 applies the models, trained on AutoViVQA only, to ViVQA-X [9] and ViVQA [30] without further training. For each dataset, we draw three random samples of about 1,000 test questions (seeds 42, 123, 3407) and score every model



Fig. 4: Validation examples. GT: one of the five references; Ours: ViBridge-VQA; Ours+LoRA: ViBridge-VQA + LoRA.

Table 6: Cross-dataset results (mean ± std over three test samples of about 1,000 questions, 510 for ViVQA; one training seed). Best per dataset in bold. All values ×100.

Model	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
ViVQA								
Vintern-1B, zero-shot	0.00 ±.00	16.01 ±.18	**60.80** ±1.17	23.90 ±.24	0.27 ±.02	15.78 ±.25	17.84 ±.19	78.63 ±2.07
Global-only + LoRA	7.38 ±.53	31.67 ±.97	34.01 ±1.58	31.48 ±1.15	0.25 ±.09	31.48 ±1.16	21.92 ±.82	84.94 ±2.87
ViBridge-VQA	16.59 ±1.48	40.59 ±1.37	48.86 ±1.89	42.68 ±1.52	0.37 ±.00	42.65 ±1.55	34.62 ±1.37	171.21 ±6.06
ViBridge-VQA + LoRA	**18.93** ±1.37	**43.08** ±.93	50.88 ±1.71	**44.76** ±1.21	**0.63** ±.02	**44.74** ±1.22	**36.98** ±1.28	**183.60** ±7.17
ViVQA-X								
Vintern-1B, zero-shot	0.20 ±.08	9.13 ±.35	**61.15** ±.80	14.52 ±.47	0.06 ±.01	8.98 ±.32	11.96 ±.41	50.13 ±2.83
Global-only + LoRA	2.57 ±.09	13.19 ±.49	26.80 ±1.24	15.89 ±.65	0.18 ±.02	11.94 ±.49	9.08 ±.57	42.12 ±2.92
ViBridge-VQA	7.40 ±.28	20.68 ±.16	41.32 ±.26	24.91 ±.14	**0.34** ±.12	19.38 ±.25	16.00 ±.21	79.51 ±1.15
ViBridge-VQA + LoRA	**8.11** ±.33	**23.07** ±.18	44.57 ±.06	**27.64** ±.11	0.33 ±.05	**21.98** ±.13	**19.24** ±.14	**95.29** ±1.51

on the same questions. Zero-shot Vintern-1B uses six tiles. Even without LoRA, ViBridge-VQA outperforms Global-only + LoRA and the zero-shot Vintern-1B

(F1 24.91 versus 15.89 and 14.52 on ViVQA-X; 42.68 versus 31.48 and 23.90 on ViVQA), and LoRA raises its F1 to 27.64 and 44.76. Zero-shot Vintern-1B keeps the highest Recall only by answering in long sentences, hence its near-zero Accuracy.

5 Conclusion

We introduced **ViBridge-VQA**, which improves Vintern-1B for generative Vietnamese VQA without updating any of its pre-trained weights. For a VLM whose projector is already aligned with its decoder, our results support three findings. First, reusing this projector with a few pooled local tokens is more effective than learning a new connector from limited data. Second, the number of visual tokens can be chosen with an explicit accuracy–cost criterion. Third, local tokens complement decoder adaptation rather than replace it: ViBridge-VQA with a decoder LoRA gives the best result.

Limitations remain: we train on a single dataset, where the task-specific ViMoE-VQA still reaches a higher F1; without decoder adaptation, exact-match accuracy stays below that of the fine-tuned Vintern-1B; and accuracy has not saturated at the 144 local tokens of our resolution.

Overall, an existing Vietnamese VLM can be improved substantially by choosing which visual tokens its frozen decoder receives, and how many. The code and the image-disjoint split are available at `https://github.com/nguyennn263/modular-vlm-finetune`.

References

1. Alayrac, J.B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al.: Flamingo: a visual language model for few-shot learning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 23716–23736 (2022)
2. Banerjee, S., Lavie, A.: METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization. pp. 65–72 (2005)
3. Cai, M., Yang, J., Gao, J., Lee, Y.J.: Matryoshka multimodal models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2025)
4. Cha, J., Kang, W., Mun, J., Roh, B.: Honeybee: Locality-enhanced projector for multimodal LLM. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13817–13827 (2024)
5. Chen, L., Zhao, H., Liu, T., Bai, S., Lin, J., Zhou, C., Chang, B.: An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In: Proceedings of the European Conference on Computer Vision (ECCV). Lecture Notes in Computer Science, vol. 15139, pp. 19–35. Springer (2024)
6. Chen, Z., Wang, W., Tian, H., Ye, S., Gao, Z., Cui, E., Tong, W., Hu, K., Luo, J., Ma, Z., et al.: How far are we to GPT-4V? Closing the gap to commercial multimodal models with open-source suites. Science China Information Sciences **67**(12), 220101 (2024)

7. Chen, Z., Wu, J., Wang, W., Su, W., Chen, G., Xing, S., Zhong, M., Zhang, Q., Zhu, X., Lu, L., Li, B., Luo, P., Lu, T., Qiao, Y., Dai, J.: InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 24185–24198 (2024)
8. Doan, K.T., Huynh, B.G., Hoang, D.T., Pham, T.D., Pham, N.H., Nguyen, Q.T.M., Vo, B.Q., Hoang, S.N.: Vintern-1B: An efficient multimodal large language model for Vietnamese (2024), arXiv preprint arXiv:2408.12480
9. Duong, T.B., Tran, H.M., Le-Nguyen, B.N., Duong, D.T.: An automated pipeline for constructing a Vietnamese VQA-NLE dataset. In: Proceedings of the International Conference on Intelligent Systems and Networks (ICISN). Lecture Notes in Networks and Systems, vol. 1596, pp. 164–173. Springer (2026)
10. Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S.: Parameter-efficient transfer learning for NLP. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 2790–2799 (2019)
11. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2022)
12. Hwang, C.L., Yoon, K.: Multiple Attribute Decision Making: Methods and Applications. Springer (1981)
13. Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 19730–19742 (2023)
14. Li, K.Y., Goyal, S., Semedo, J.D., Kolter, J.Z.: Inference optimal VLMs need fewer visual tokens and more parameters. In: Proceedings of the International Conference on Learning Representations (ICLR) (2025)
15. Li, W., Yuan, Y., Liu, J., Tang, D., Wang, S., Qin, J., Zhu, J., Zhang, L.: Token-Packer: Efficient visual projector for multimodal LLM. International Journal of Computer Vision **133**(10), 6794–6812 (2025)
16. Li, X.L., Liang, P.: Prefix-tuning: Optimizing continuous prompts for generation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 4582–4597 (2021)
17. Lin, C.Y.: ROUGE: A package for automatic evaluation of summaries. In: Proceedings of the ACL Workshop on Text Summarization Branches Out. pp. 74–81 (2004)
18. Lin, T.Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., Perona, P., Ramanan, D., Zitnick, C.L., Dollár, P.: Microsoft COCO: Common objects in context. In: Proceedings of the European Conference on Computer Vision (ECCV). pp. 740–755 (2014)
19. Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 26296–26306 (2024)
20. Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 34892–34916 (2023)
21. Nguyen, A.T., Nguyen, T.Q., Dang, D.L., Tran, D.T., Phan, B.D., Nguyen, Q.T., Le, T.: ViMoE-VQA: Reasoning-aware sparse experts for generative Vietnamese visual question answering. Procedia Computer Science (2026), to appear

22. Nguyen, A.T., Phan, B.D., Nguyen, T.Q., Tran, D.T., Dang, D.L., Nguyen, Q.T., Le, T.: AutoViVQA: A large-scale, automatically constructed dataset for Vietnamese visual question answering. In: Proceedings of the Asian Conference on Intelligent Information and Database Systems (ACIIDS). Communications in Computer and Information Science, vol. 2964, pp. 492–506. Springer (2026)
23. Nguyen, N.S., Nguyen, V.S., Le, T.: Advancing Vietnamese visual question answering with transformer and convolutional integration. Computers and Electrical Engineering **119**, 109474 (2024)
24. Nguyen, Q.V., Tran, D.Q., Pham, H.Q., Nguyen, T.K.B., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViTextVQA: A large-scale visual question answering dataset and a novel multimodal feature fusion method for Vietnamese text comprehension in images. Expert Systems with Applications **308**, 130839 (2026)
25. Papineni, K., Roukos, S., Ward, T., Zhu, W.J.: BLEU: a method for automatic evaluation of machine translation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 311–318 (2002)
26. Pham, H.Q., Nguyen, T.K.B., Nguyen, Q.V., Tran, D.Q., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViOCRVQA: Novel benchmark dataset and VisionReader for visual question answering by understanding Vietnamese text in images. Multimedia Systems **31**(2), 106 (2025)
27. Qwen Team: Qwen2.5 technical report (2024), arXiv preprint arXiv:2412.15115
28. Satopää, V., Albrecht, J., Irwin, D., Raghavan, B.: Finding a "kneedle" in a haystack: Detecting knee points in system behavior. In: Proceedings of the International Conference on Distributed Computing Systems Workshops (ICDCSW). pp. 166–171 (2011)
29. Shang, Y., Cai, M., Xu, B., Lee, Y.J., Yan, Y.: LLaVA-PruMerge: Adaptive token reduction for efficient large multimodal models. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 22857–22867 (2025)
30. Tran, K.Q., Nguyen, A.T., Le, A.T.H., Nguyen, K.V.: ViVQA: Vietnamese visual question answering. In: Proceedings of the Pacific Asia Conference on Language, Information and Computation (PACLIC). pp. 683–691 (2021)
31. Tran, K.V., Nguyen, K.V., Nguyen, N.L.T.: BARTPhoBEiT: Pre-trained sequence-to-sequence and image transformers models for Vietnamese visual question answering. In: Proceedings of the International Conference on Multimedia Analysis and Pattern Recognition (MAPR). pp. 1–6 (2023)
32. Tran, K.V., Phan, H.P., Nguyen, K.V., Nguyen, N.L.T.: ViCLEVR: A visual reasoning dataset and hybrid multimodal fusion model for visual question answering in Vietnamese. Multimedia Systems **30**(4), 199 (2024)
33. Tsimpoukelli, M., Menick, J., Cabi, S., Eslami, S.M.A., Vinyals, O., Hill, F.: Multimodal few-shot learning with frozen language models. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 200–212 (2021)
34. Vedantam, R., Zitnick, C.L., Parikh, D.: CIDEr: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4566–4575 (2015)
35. Yang, S., Chen, Y., Tian, Z., Wang, C., Li, J., Yu, B., Jia, J.: VisionZip: Longer is better but not necessary in vision language models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 19792–19802 (2025)
36. Yao, L., Li, L., Ren, S., Wang, L., Liu, Y., Sun, X., Hou, L.: DeCo: Decoupling token compression from semantic abstraction in multimodal large language models (2024), arXiv preprint arXiv:2405.20985